Comprehensive Hyperparameter & System Specification Tables

The following table details all structural, architectural, and algorithmic hyperparameters governing TraceScope AI 2.0:

Model / Pipeline Componen...	Hyperparameter / Setting	Value / Specification	Rationale / Standard
Tier 1: Random Forest Baseli...	Number of Estimators (Trees)	100	Ensemble variance reduction
Tier 1: Random Forest Baseli...	Split Criterion	Information Gain (Entropy)	Maximal discrimination on sta...
Tier 1: Random Forest Baseli...	Max Tree Depth	None (Full Expansion)	Leaf purity bound
Tier 1: Random Forest Baseli...	Random Seed	42	Deterministic reproducibility
Tier 1: Random Forest Baseli...	Input Feature Dimensions	10 Features	Statistical noise moments & e...
Tier 1: SVM Baseline	Kernel Function	Radial Basis Function (RBF)	Nonlinear boundary projectio...
Tier 1: SVM Baseline	Regularization Parameter ©	10.0	Balanced margin violation pe...
Tier 1: SVM Baseline	Kernel Coefficient (gamma)	scale (1 / (n_features * X.var(...	Variance-adaptive bandwidth
Tier 1: SVM Baseline	Feature Scaling	StandardScaler (Zero-mean, ...	RBF isotropic metric requirem...
Tier 2: Deep ResNet-18 Base...	Base Backbone	PyTorch torchvision ResNet-1...	Standard residual benchmark
Tier 2: Deep ResNet-18 Base...	Forensic Pre-filter	Kraetzer-Vogler 5x5 High-Pas...	Suppress typographic docum...
Tier 2: Deep ResNet-18 Base...	Input Patch Dimensions	256 x 256 x 1 (Grayscale Res...	High-frequency spatial resolut...
Tier 2: Deep ResNet-18 Base...	Optimizer & Initial LR	Adam (Initial LR = 1e-4, beta...	Adaptive momentum gradient...
Tier 2: Deep ResNet-18 Base...	Weight Decay	1e-4	L2 regularization against over...
Tier 2: Deep ResNet-18 Base...	Batch Size & Epochs	Batch Size = 32, Epochs = 25	GPU memory & stable conver...
Tier 2: Deep ResNet-18 Base...	LR Scheduler	ReduceLROnPlateau (factor=...	Fine-grained local minimum d...
Tier 3: Flagship Hybrid CNN	Spatial Branch Architecture	3x Conv2D (32, 64, 128 filters...	Deep spatial noise descriptor ...
Tier 3: Flagship Hybrid CNN	Handcrafted Branch Architect...	Dense(64, ReLU) + Dropout(...	Physical feature dimension ali...
Tier 3: Flagship Hybrid CNN	Fusion & Bottleneck Layers	Concatenate(192) -> Dense(2...	Cross-modal late fusion bottle...
Tier 3: Flagship Hybrid CNN	Handcrafted Descriptors	44 Dimensions (11 PRNU + 3...	Multi-domain physical fingerpr...
Tier 3: Flagship Hybrid CNN	Optimizer & Batch Size	Adam (LR = 1e-4), Batch Siz...	Loss convergence without gra...
Tier 3: Flagship Hybrid CNN	Loss Function	Categorical Cross-Entropy	Multi-class probabilistic attrib...
Open-Set Rogue Rejection M...	Distance Metric & Latent Spa...	Euclidean Distance to Known ...	Direct feature geometry witho...
Open-Set Rogue Rejection M...	Calibrated Decision Threshol...	tau = 2.84	Yields 95.16% Unknown TPR...
Tampering Localization Engin...	Sliding Window Patch Size & ...	Window = 64 x 64 pixels, Stri...	Sub-word typographic bound...
Tampering Localization Engin...	Residual Variance Ratio Thre...	V_ratio < 0.35 (Inpainting) or ...	Energy deficit & boundary ste...